

Profile Summary

Machine Learning Engineer with hands-on experience building production-style ML systems using Scikit-Learn, FastAPI, Docker, and AWS S3. Skilled in designing modular pipelines for data ingestion, preprocessing, model training, evaluation, and deployment, with a strong software engineering foundation for building scalable AI applications.

Skills

- **Languages:** Python, SQL, JavaScript
- **Machine Learning:** Scikit-Learn, Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Data:** Pandas, Numpy, Matplotlib, Seaborn
- **MLOps & Deployment:** FastAPI, Docker, AWS S3
- **Software Engineering:** Git, GitHub, ML Pipelines, Logging, Exception Handling, Configuration Management

EXPERIENCE

360DigiTMG — Data Analyst Intern | Feb 2024 – Apr 2024

- Performed data cleaning, preprocessing, and exploratory data analysis using Python, Pandas, and NumPy.
- Created visualizations using Matplotlib and Seaborn to identify user behavior patterns and trends.
- Analyzed website engagement metrics using SQL queries and Python to identify user behavior patterns.
- Generated actionable insights and presented findings through reports and visual dashboards.

Slashmark — Machine Learning Intern | Dec 2023 – Feb 2024

- Built and evaluated supervised and unsupervised machine learning models using Scikit-Learn.
- Performed data preprocessing, feature selection, and model tuning to improve performance.
- Applied Linear Regression, Logistic Regression, and K-Means clustering to real-world datasets.
- Evaluated models using Accuracy, Precision, Recall, F1-Score, and RMSE metrics.

PROJECTS

Vehicle Insurance Prediction System | [GitHub Link](#) | Python | Scikit-Learn | FastAPI | Docker | AWS S3

- Built a production-style end-to-end machine learning system for predicting vehicle insurance purchase likelihood.
- Implemented modular components for **Data Ingestion, Data Validation, Data Transformation, Model Training, Model Evaluation, and Prediction Pipeline**.
- Automated preprocessing using Scikit-Learn Pipelines with feature engineering, missing value handling, categorical encoding, and scaling.
- Containerized the application using **Docker** for reproducible deployment.
- Integrated **Amazon S3** for storing datasets, trained models, and pipeline artifacts to support reproducible training and deployment.
- Exposed prediction APIs using **FastAPI**, enabling real-time inference.
- Applied logging, exception handling, configuration management, and artifact versioning to mirror production ML workflows.

End to End House Price Prediction | [GitHub Link](#) | Python | Scikit-Learn | FastAPI | Modular ML Pipeline

- Designed and implemented an end-to-end machine learning pipeline using a modular architecture with separate components for **Data Ingestion, Data Transformation, Model Training, Prediction Pipeline, Logging, and Exception Handling**.
- Built reusable preprocessing pipelines using **ColumnTransformer, Pipeline, SimpleImputer, OneHotEncoder**, and **StandardScaler**.
- Compared **Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost** using **RMSE** to select the best-performing regression model.
- Developed a FastAPI inference service for real-time house price prediction and serialized trained models using Joblib/Pickle.
- Followed production-style project structure with artifacts management, configuration classes, and reusable utility modules.

Dog Breed Classification System | [GitHub Link](#) | TensorFlow, Keras, Flask

- Fine-tuned Xception, Inception, and DenseNet models using transfer learning on the Stanford Dogs dataset.
- Built an ensemble inference pipeline to improve prediction robustness and overall accuracy.
- Implemented a complete training, evaluation, and validation pipeline using accuracy, F1-score, and confusion matrix.
- Deployed the model using a Flask API that accepts image uploads and returns predicted dog breeds.

EDUCATION

B.Tech in Computer Science and Engineering
Narasaraopeta Engineering College, India (2020 – 2024) | CGPA: **7.96**

CERTIFICATIONS

- Machine Learning Specialization — Coursera (Andrew Ng)
- The Joy of Computing using Python — NPTEL